

EE326 Intro. to Information Theory and Coding Midterm

Spring 2024

Date: Monday, Apr. 15, 2024

Time: 9am-11:30am

Problem 1 (5 points)

Below you have five True/False questions. If you write a correct answer, you get 1 point for each question, but if you write an incorrect answer, you get -1 point for each question. If you don't write any answer, you get 0 point. (*You don't need to write down your reasonings for T/F questions.*)

- a) For any $(X, Y, Z) \sim P_{XYZ}$, it is ALWAYS true that for any $z \in \mathcal{Z}$

$$I(X; Y) \geq I(X; Y | Z = z).$$

True or False?

- b) For any $(X, Y, Z) \sim P_{XYZ}$, it is ALWAYS true that

$$I(X; Y | Z) \geq 0.$$

True or False?

- c) For any $(X, Y) \sim P_{XY}$, it is ALWAYS true that

$$H(X) + H(Y) - 2I(X; Y) = 2H(X, Y) - H(X) - H(Y)$$

True or False?

- d) For any random variable $(X, Y) \sim P_{XY}$ where $X \in \mathcal{X}$ for a finite $|\mathcal{X}|$ and any function $g : \mathcal{X} \rightarrow \mathbb{R}$, it is ALWAYS true that

$$I(g(X); Y) \geq I(X; Y).$$

True or False?

- e) For any $(X, Y, Z) \sim P_{XYZ}$ forming a Markov chain, $X \rightarrow Y \rightarrow Z$,

$$D(P_{XYZ} \| [P_X \cdot P_Y]) = I(X; Y)$$

where P_X and P_Y are marginal distributions of P_{XYZ} . True or False?

Problem 2 (Write down your reasoning to get the full score: 15 points)

In this problem, we will see that information theory can provide some intuition in designing gambling strategies. Consider the following formulation. At each time $t = 1, 2, \dots$, a gambler has a finite set of choices that he/she can make. For example, think of it as K horses in a race, when only one of them can win in the race. We model the horse that wins at time t as a random variable $X_t \in \{1, 2, \dots, K\}$ with probability distribution $[p_1, \dots, p_K]$, i.e., $\Pr(X_t = k) = p_k$. If one bets 1 dollar on the k -th horse, then he wins r_k dollars if $X_t = k$ and loses the dollar if any other horse wins. Here, r_k is called the payback ratio.

The gambler starts at time $t = 1$, with some amount of money, say $M_0 = 1$ dollar, and he/she needs to come up with an allocation strategy over the horses (a nicer word is portfolio). Suppose that at each time, the gambler invests b_k fraction of all the money he/she has, including the winning money from the previous games, on horse k . Clearly $b_k \geq 0$ and $\sum_{k=1}^K b_k = 1$. We assume that the allocation vector $B = [b_1, \dots, b_K]$, payback ratio $R = [r_1, \dots, r_K]$, and the distribution of games $P_X = [p_1, \dots, p_K]$ are time invariant, i.e., fixed over all time. It is easy to verify that at time n , the gambler's wealth is

$$M_n = M_0 \cdot \prod_{t=1}^n \left(\sum_{k=1}^K b_k r_k \mathbb{1}(X_t = k) \right)$$

where $\mathbb{1}(X_t = k) = 1$ if $X_t = k$ and $\mathbb{1}(X_t = k) = 0$ otherwise. Thus, we can define the exponential growth rate of the gambler's wealth as

$$W(B, P_X, R) := \mathbb{E}_{P_X} \left[\log \left(\sum_{k=1}^K b_k r_k \mathbb{1}(X_t = k) \right) \right] = \sum_{k=1}^K p_k \log(b_k r_k).$$

(Write down your reasonings (or explicit calculations) for each question to get the full score.)

a) (2 points) First, show that

$$W(B, P_X, R) = \sum_{k=1}^K p_k \log r_k - H(P_X) - D(P_X \| B).$$

b) (3 points) What is the optimal betting strategy B^* to maximize $W(B, P_X, R)$?

c) (5 points) Of course, the problem is that no one knows the exact P_X , precisely. Suppose that the casino owner has an estimate of the distribution P_X denoted by $P'_X = [p'_1, \dots, p'_K]$, and sets the payback ratios to be $r_k = 1/p'_k$. Then, write down $W(B, P_X, R)$ as a function of $D(P_X \| P'_X)$ and $D(P_X \| B)$.

d) (5 points) Now, suppose that there is some side information, modeled as another random variable Y_t , jointly distributed with X_t , with distribution P_{XY} , where (X_t, Y_t) is i.i.d. over time. A gambler who observes the side information $Y = y \in \mathcal{Y}$ would use an allocation $B_{\{Y=y\}} = P_{X|Y}(\cdot|y)$, assuming $P_{X|Y}(\cdot|y)$ is known to the gambler. Let the long-term average exponential growth rate with the information

be denoted by W' :

$$\begin{aligned} W'(B, P_{XY}, R) &= \mathbb{E}_{P_{XY}} \left[\log \left(\sum_{k=1}^K b_k r_k \mathbb{1}(X_t = k) \right) \right] \\ &= \sum_{y \in \mathcal{Y}} P_Y(y) \sum_{k=1}^K P_{X|Y}(k|y) \log(b_k r_k). \end{aligned}$$

Show the connection between

$$W'(B^{**}, P_{XY}, R) - W(B^*, P, R)$$

and $I(X; Y)$ where $B^{**} = B_{\{Y=y\}} = [P_{X|Y}(1|y), \dots, P_{X|Y}(K|y)]$ and B^* is the allocation maximizing $W'(B, P_X, R)$ (your solution of problem b)). Briefly discuss the meaning of your result based on the sign of $W'(B^{**}, P_{XY}, R) - W(B^*, P, R)$.

Problem 3 (Write down your reasoning to get the full score: 10 points)

Let P_{XY} be a given joint distribution of X and Y , defined over alphabets $\mathcal{X}$ and $\mathcal{Y}$, respectively. We write P_X and P_Y as the corresponding marginal distributions of X and Y , respectively. The two subproblems a) and b) below are independent.

- a) (5 points) In class, we defined the type and strong typical set, with respect to a distribution of a single random variable. In this problem, we generalize this to joint distributions and consider joint type and strong joint typical set.

A pair of sequences (x^n, y^n) is said to have joint type of P_{XY} if its joint empirical distribution is P_{XY} , i.e.,

$$\begin{aligned}\hat{P}_{(x^n, y^n)}(a, b) &\triangleq \frac{1}{n} \sum_{i=1}^n \mathbb{1}((x_i, y_i) = (a, b)) = \frac{1}{n} N((a, b) | (x^n, y^n)) \\ &= P_{XY}(a, b), \quad \forall a \in \mathcal{X}, \forall b \in \mathcal{Y}.\end{aligned}$$

Let's define the strong joint typical sets with respect to P_{XY} , P_X and P_Y , respectively, as follows:

$$\tilde{T}_{P_{XY}} = \{(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n : |\hat{P}_{(x^n, y^n)}(a, b) - P_{XY}(a, b)| \leq \delta, \quad \forall a \in \mathcal{X}, \forall b \in \mathcal{Y}\}, \quad (1)$$

$$\tilde{T}_{P_X} = \{x^n \in \mathcal{X}^n : |\hat{P}_{x^n}(a) - P_X(a)| \leq C_1 \cdot \delta, \quad \forall a \in \mathcal{X}\}, \quad (2)$$

$$\tilde{T}_{P_Y} = \{y^n \in \mathcal{Y}^n : |\hat{P}_{y^n}(b) - P_Y(b)| \leq C_2 \cdot \delta, \quad \forall b \in \mathcal{Y}\}, \quad (3)$$

for some constants $C_1, C_2 > 0$ and $\delta > 0$. I ask you to prove the following statement:

$$\text{If } (x^n, y^n) \in \tilde{T}_{P_{XY}}, \text{ then } x^n \in \tilde{T}_{P_X} \text{ and } y^n \in \tilde{T}_{P_Y}.$$

Find two proper constants $C_1 > 0$ in (2) and $C_2 > 0$ in (3), to make this statement true for any $\delta > 0$. Assume that $|\mathcal{X}|$ and $|\mathcal{Y}|$ are finite.

- b) (5 points) Given a sequence of samples $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, we want to compute and compare two probabilities (likelihoods) 1) $\prod_{i=1}^n P_X(x_i)P_Y(y_i)$ (probability of x_i 's and y_i 's being generated independently by P_X and P_Y , respectively.) and 2) $\prod_{i=1}^n P_{XY}(x_i, y_i)$ (probability of (x_i, y_i) 's being generated by the joint distribution $P_{XY}(x, y)$). If the probability 1) is larger than 2), we would believe that the samples x_i 's and y_i 's are generated independently by $P_X(x)$ and $P_Y(y)$, respectively. On the other hand, if the probability 2) is larger than 1), then we would believe that the pairs of samples (x_i, y_i) 's are generated by the joint distribution $P_{XY}(x, y)$. This kind of test is called *likelihood ratio test*.

Let's assume that the samples are generated by the joint distribution P_{XY} , i.e., $(X_i, Y_i) \sim \text{i.i.d. } P_{XY}$. Find the limit

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \frac{(\prod_{i=1}^n P_X(X_i)P_Y(Y_i))}{(\prod_{i=1}^n P_{XY}(X_i, Y_i))}$$

that holds with high probability. From your answer, discuss how small $\prod_{i=1}^n P_X(x_i)P_Y(y_i)$ is compared to $\prod_{i=1}^n P_{XY}(x, y)$, when the true distribution of the samples was $P_{XY}(x, y)$.